



Assignment-1

Data Lakehouse with Snowflake

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Table of Contents

1.	Introduction	2
a.	Part 1: Data Ingestion	2
b.	Part 2: Data Cleaning	6
c.	Part 3: Data Analysis	10
d.	Part 4: Business Question	13
	Conclusion	18
	References	18

1. Introduction

The objective of this assignment is to analyse a YouTube Trending dataset using a Data Lakehouse built on Snowflakes. The data set contains top daily popular videos trending in 10 countries like India, USA, Great Britain, Germany, Canada, France, Brazil, Mexico, South Korea, and Japan for the time duration from 12-Aug-2020 to 15-April-2024.

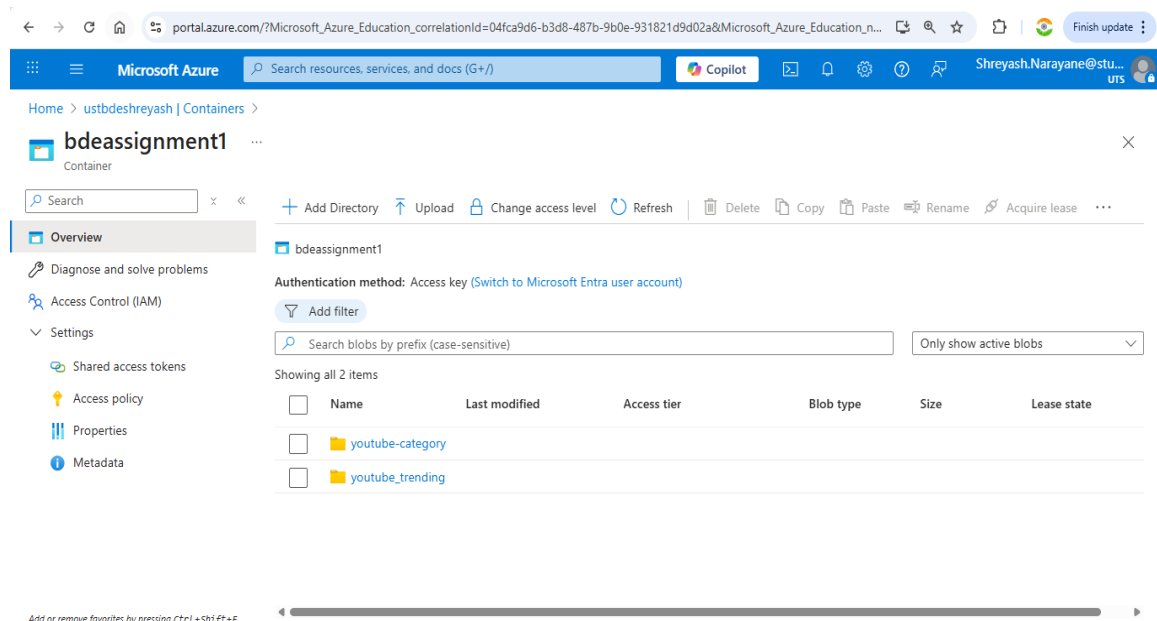
a. Part 1: Data Ingestion

1. Download the (compressed) dataset on:

The YouTube Trending Videos compressed dataset was downloaded from the google drive link provided in the assignment file. This dataset contains Trending data in CSV format and Category data in JSON format.

2. Upload the dataset in your storage account on Azure

At first, an account was created on Azure. New storage account was created named “ustbdeshtreyash” to store the data. And in this storage account, a container was created named “bdeassignment1”. The dataset which was downloaded in the previous, all the CSV files and JSON files were uploaded to this container named “bdeassignment1”.



Here, after uploading the Azure SAS token was generated. This token will be used to set up the connection of Azure storage with Snowflakes. An external storage in Snowflake will be created to fetch the data from Azure. The data is secured and will only be shared only for a limited time and will be read-only access for any users having access to it.

3. On Snowflake:

- i. A Snowflake account was created using university email. Here, a new database was created named "assignment_1" as below:

```
1 CREATE OR REPLACE DATABASE assignment_1;
2 --Database created
3
4 USE DATABASE assignment_1;
```

Results

status
1 Database ASSIGNMENT_1 successfully created.

```
4 USE DATABASE assignment_1;
```

Results

status
1 Statement executed successfully.

- ii. Then, a stage called "**stage_assignment**" was created in Snowflakes which will connect to Azure Blob Storage container using a secure SAS token generator earlier.

```
9 --Creating stage with azure url and azure SAS token
10 CREATE OR REPLACE STAGE stage_assignment
11 URL='azure://ustbdeshtreyash.blob.core.windows.net/bdeassignment1'
12 CREDENTIALS=(AZURE_SAS_TOKEN='sv=2024-11-04&ss=b&srt=co&sp=rwdlacietyfx&se=2026-08-19T12:27:25Z&st=2025-08-19T04:12:25Z&spr=https&sig=aysFm4ZLtxaPrH12g3XguhZ%2B25AhzEHGFpXSYL%2BYHW0%3D')
13 ;
14
```

Results

status
1 Stage area STAGE_ASSIGNMENT successfully created.

```
15 list @stage_assignment;
16 --listing stage
17
```

Results

	name	size	md5	last_modified
1	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	6148	194577a7e20bdcc7afbb718f502c134c	Mon, 18 Aug 2025 10:27:57 GMT
2	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
3	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
4	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
5	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
6	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
7	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
8	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT
9	azure://ustbdeshtreyash.blob.core.windows.net/bdeassignmen	10096	e5f8d4189665ae35616bd65ff261c42b	Mon, 18 Aug 2025 10:27:57 GMT

4. Ingest the data as external tables on Snowflake

One external table is created for Trending data CSV file named, **“ex_table_youtube_trending”**. Another external table is created for JSON files containing Category data, which is named **“ex_table_youtube_category”**.

```
CREATE OR REPLACE EXTERNAL TABLE ex_table_youtube_trending (
  video_id      STRING      AS (VALUE:c1::STRING),
  title         STRING      AS (VALUE:c2::STRING),
  publishedat   TIMESTAMP_NTZ AS (TO_TIMESTAMP_NTZ(VALUE:c3::STRING)),
  channelid     STRING      AS (VALUE:c4::STRING),
  channeltitle  STRING      AS (VALUE:c5::STRING),
  categoryid    NUMBER       AS (TRY_TO_NUMBER(VALUE:c6::STRING)),
  trending_date DATE        AS (TO_DATE(VALUE:c7::STRING)),
  view_count    NUMBER       AS (TRY_TO_NUMBER(VALUE:c8::STRING)),
  likes         NUMBER       AS (TRY_TO_NUMBER(VALUE:c9::STRING)),
  dislikes      NUMBER       AS (TRY_TO_NUMBER(VALUE:c10::STRING)),
  comment_count NUMBER       AS (TRY_TO_NUMBER(VALUE:c11::STRING)),
  country       STRING      AS (
    UPPER(REGEXP_SUBSTR(METADATA$FILENAME, '^[^/]*([A-Za-z]{2})_', 1, 1, 'e', 2))
  )
WITH LOCATION = @stage_assignment
FILE_FORMAT = (
  TYPE = CSV
  FIELD_DELIMITER = ','
  SKIP_HEADER = 1
  FIELD_OPTIONALLY_ENCLOSED_BY = ''
  TRIM_SPACE = TRUE
  NULL_IF = ('', 'NULL', 'null')
) PATTERN = '.*\\.csv$'
AUTO_REFRESH = FALSE;
```

```
--fetching rowcount to ensure the table is created
SELECT COUNT(1) FROM ex_table_youtube_trending;
```

results		Chart
# COUNT(1)		
		2667041

```
-- External table on the JSON files
CREATE OR REPLACE EXTERNAL TABLE ex_table_youtube_category (raw VARIANT AS (value))
WITH LOCATION = @stage_assignment
FILE_FORMAT = (TYPE = JSON)
PATTERN = '.*_category_id\\.json$'
AUTO_REFRESH = FALSE;

--fetching rowcount to ensure the table is created:
SELECT count(1) FROM ex_table_youtube_category;
```

Results		Chart
# COUNT(1)		
		10

5. Transfer the data from external tables into tables

- i. A new table is created called “**table_youtube_trending**”, to store the data from “ex_table_youtube_trending”. This will save all the records required and present in the external table, which will eventually save time, as the fetching speed will increase and performance will be much better.

```
--creating new table table_youtube_trending from ex_table_youtube_trending
CREATE OR REPLACE TABLE table_youtube_trending AS
SELECT
    video_id,
    title,
    channelid,
    channeltitle,
    publishedat,
    trending_date,
    view_count,
    likes,
    dislikes,
    comment_count,
    categoryid,
    country
FROM ex_table_youtube_trending;

--fetching rowcount to ensure the table is created:
SELECT COUNT(1) AS rows_in_table_trending FROM table_youtube_trending;
```

Results	
Chart	
# ROWS_IN_TABLE_TRENDING	
	2667041

- ii. Another table is created but now for category data, named called “**table_youtube_category**”. Here the JSON data is transformed into a structured table in Snowflakes. New column “country” is created and was extracted from the file name.

```
--creating new table table_youtube_category from ex_table_youtube_category
CREATE OR REPLACE TABLE table_youtube_category AS
SELECT DISTINCT
    UPPER(REGEXP_SUBSTR(METADATA$FILENAME, '([A-Za-z]{2})_', 1, 1, 'e', 1)) AS country,
    TRY_TO_NUMBER(item.value:id::string) AS category_id,
    item.value:snippet:title::string AS category_title
FROM ex_table_youtube_category,
    LATERAL FLATTEN(input => raw:items) AS item;
```

Results	
Chart	
status	
Table TABLE_YOUTUBE_CATEGORY successfully created.	

6. A final table is created, called “**table_youtube_final**” by combining (joining) “table_youtube_trending” with “table_youtube_category” on *country* and *categoryid* columns. Additionally, added a new field called “**id**” by using the “UUID_STRING()” function.

```
--merging table_youtube_category and table_youtube_trending to create a final table called : table_youtube_final
--new column name : 'id' added
CREATE OR REPLACE TABLE table_youtube_final AS
SELECT UUID_STRING() AS id,
       t.video_id,
       t.title,
       t.channelid,
       t.channeltitle,
       t.publishedat,
       t.trending_date,
       t.view_count,
       t.likes,
       t.dislikes,
       t.comment_count,
       t.categoryid,
       t.country,
       c.category_title
FROM table_youtube_trending t
LEFT JOIN table_youtube_category c
  ON t.country = c.country
  AND t.categoryid = c.category_id;

-- Checking count
SELECT COUNT(1) AS final_row_count FROM table_youtube_final;
--expected count 2,667,041
-----
-----
```

Results ~ Chart

FINAL_ROW_COUNT

2667041

The count of **2,667,041** rows was fetched from the table “table_youtube_final.”

b. Part 2: Data Cleaning

1. The category_title having duplicates in the table “table_youtube_category” (not considering categoryid) : “Comedy”

```
1  USE DATABASE assignment_1;
2
3  --1.) category_title having duplicates in “table_youtube_category”
4  SELECT category_title
5  FROM table_youtube_category
6  GROUP BY category_title
7  HAVING COUNT(DISTINCT category_id) > 1;
8
```

Results ~ Chart

A CATEGORY_TITLE

Comedy

2. The category_title which appears only in one country in the table “table_youtube_final”:
“Nonprofits & Activism”

```

12 --2.) category_title appearing only in one country
13 SELECT category_title, country
14 FROM (SELECT category_title,
15         COUNT(DISTINCT country) AS country_count,
16         MIN(country) AS country
17        FROM table_youtube_category
18        GROUP BY category_title
19       ) AS iq
20 WHERE country_count = 1;
21
22 -----
23 -----

```

→ Results	Chart
⌵ CATEGORY_TITLE	⌵ COUNTRY
Nonprofits & Activism	US

3. The categoryid of the missing category_titles in the table “table_youtube_final” : “29”

```

25 --3.) categoryid of the missing category_titles
26
27 SELECT DISTINCT categoryid
28 FROM table_youtube_final
29 WHERE category_title IS NULL OR category_title = '';
30

```

→ Results	Chart
# CATEGORYID	
	29

4. Update category_title column in the table “table_youtube_final”: replace the NULL values in category_title with the answer from the previous question.

```

47 --Update statement:
48 UPDATE table_youtube_final
49 SET category_title = (SELECT category_title
50                     FROM table_youtube_category
51                     WHERE category_id=(SELECT DISTINCT categoryid
52                                       FROM table_youtube_final
53                                       WHERE category_title IS NULL OR category_title = ''))
54 WHERE category_title IS NULL OR category_title = '';
55
56

```

→ Results	Chart
# number of rows updated	# number of multi-joined rows updated
1	1563
	0

5. Video which does not have a “channeltitle” in the table “table_youtube_final” :
“Kala Official Teaser | Tovino Thomas | Rohith V S | Juvis Productions | Adventure Company”

```
70 --5.) Video not having a channeltitle
71 SELECT title
72 FROM table_youtube_final
73 WHERE channeltitle IS NULL OR channeltitle = '';
74
75 -----
```

→ Results Chart

TITLE
Kala Official Teaser Tovino Thomas Rohith V S Juvis Productions Adventure Company

6. Delete records having video_id = “#NAME?”

```
78 --6.) Delete records having video_id = “#NAME?”
79 SELECT COUNT(1) FROM table_youtube_final
80 WHERE video_id = '#NAME?';
81 --32081 records with video_id = '#NAME?'
82
83 SELECT COUNT(1) FROM table_youtube_final
84 WHERE video_id <> '#NAME?';
85 --2634960 records
86
87 DELETE FROM table_youtube_final
88 WHERE video_id = '#NAME?';
89 --32081 records deleted
```

↶ Results Chart

number of rows deleted
1 32081

Records deleted: 32081

7. Create a new table called “table_youtube_duplicates” containing only the “bad” duplicates using the row_number() function.

```
111
112 CREATE OR REPLACE TABLE table_youtube_duplicates AS
113 SELECT *
114 FROM (
115     SELECT *,
116     ROW_NUMBER() OVER (
117         PARTITION BY video_id, country, trending_date
118         ORDER BY view_count DESC
119     ) AS rnum
120 FROM table_youtube_final
121 )
122 WHERE rnum > 1;
123
```

↶ Results Chart

status
1 Table TABLE_YOUTUBE_DUPLICATES successfully created.

```

125 | SELECT COUNT(1) FROM table_youtube_duplicates;
126 | --Count expected 37466
127 |
128 | -----

```

# COUNT(1)	
1	37466

The count of “bad” duplicates records = 37466

8. Delete the duplicates in “table_youtube_final” by using “table_youtube_duplicates”

```

145 | --deleting duplicate records:
146 | DELETE FROM table_youtube_final
147 | WHERE id IN (SELECT id FROM table_youtube_duplicates);
148 |

```

# number of rows deleted	
1	37466

The number of duplicates records deleted = 37466

9. Count the number of rows in “table_youtube_final” after the data cleaning = 2,597,494 rows

```

159 | --9.) Count the number of rows in “table_youtube_final” and check that it is equal to 2,597,494 rows
160 | SELECT COUNT(1) FROM table_youtube_final;
161 | --Latest count = 2597494 rows
162 |
163 | -----END-----
164 | -----

```

# COUNT(1)	
1	2597494

c. Part 3: Data Analysis

1. Three most viewed videos for each country in the 'Gaming' category where "trending_date" = "2024-04-01". Order by country and rank.

```
6  --1.) 3 most viewed videos for each country in the 'Gaming' category where trending_date = "2024-04-01". Order by country and rank
7  SELECT country, title, channeltitle, view_count,
8         ROW_NUMBER() OVER (
9             PARTITION BY country
10            ORDER BY view_count DESC
11           ) AS ROWNUM
12 FROM table_youtube_final
13 WHERE trending_date = '2024-04-01'
14    AND category_title = 'Gaming'
15    QUALIFY ROWNUM <= 3
16 ORDER BY country, ROWNUM ASC;
```

	COUNTRY	TITLE	CHANNELTITLE	VIEW_COUNT	ROWNUM
1	BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
2	BR	iShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	iShowSpeed	2971782	2
3	BR	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
4	CA	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
5	CA	If my viewers break my secret rule, I ban them	DougDoug	2988844	2

This query fetches the top three most-viewed gaming videos per country on 01-Apr-2024. To rank the videos by view count within each country, ROW_NUMBER() function is used, which is then filtered for the top three. Results are ordered by country and rank, showing Country, Channel Title, channel, and views for comparison. Results show thirty records, three records each for ten countries.

2. Distinct count of video with title containing the word "BTS" and group by country and order the result by count in a descending order.

```
21  --2.) Distinct count of video with title containing the word "BTS", group by country and order the result by count in a descending order
22  SELECT country, COUNT(DISTINCT video_id) AS CT
23 FROM table_youtube_final
24 WHERE UPPER(title) ILIKE '%BTS%'
25 GROUP BY country
26 ORDER BY ct DESC, country ASC;
```

	COUNTRY	CT
1	KR	468
2	IN	288
3	US	268
4	CA	262
5	MX	254
6	JP	251
7	DE	242
8	GB	223

Here the query counts distinct videos with "BTS" in their title for each country. While searching the word "BTS" and ILIKE is used to ensure the word "BTS" is checked inside all the rows of "title" column and "UPPER(title)" is used to fetch case-insensitive matching. All ten countries result is fetched, and highest distinct count is for South Korea, then India, then USA, Canada, Mexico and more.

- Most viewed video for each country and every month of 2024 with their “likes_ratio”, order by “year_month” and country.

```

31 --3.) Most viewed video for each country and every month of 2024 with their Likes_ratio, order by year_month and country.
32 WITH base AS (
33     SELECT country, DATE_TRUNC('month', trending_date) AS year_month, title, channeltitle, category_title, view_count, likes
34     FROM table_youtube_final
35     WHERE trending_date >= '2024-01-01'
36     AND trending_date < '2025-01-01'
37 ),
38 ranked AS (
39     SELECT b.*, ROW_NUMBER() OVER (
40         PARTITION BY country, year_month
41         ORDER BY view_count DESC
42     ) AS rn
43     FROM base b
44 )
45 SELECT country, year_month, title, channeltitle, category_title, view_count, ROUND((likes / NULLIF(view_count, 0)) * 100, 2) AS likes_ratio
46 FROM ranked
47 WHERE rn = 1
48 ORDER BY year_month, country;

```

This query finds the most-viewed video for each country and month in 2024. The ROW_NUMBER() function ranks videos monthly per country by views.

Result:

	COUNTRY	YEAR_MONTH	TITLE	CHANNELTITLE	CATEGORY_TITLE	VIEW_COUNT	LIKES_RATIO
1	BR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	139504939	3.20
2	CA	2024-01-01	Still Here Season 2024 Cinematic - League of	League of Legends	Gaming	104159411	1.69
3	DE	2024-01-01	Still Here Season 2024 Cinematic - League of	League of Legends	Gaming	104159411	1.69
4	FR	2024-01-01	Still Here Season 2024 Cinematic - League of	League of Legends	Gaming	104159411	1.69
5	GB	2024-01-01	Still Here Season 2024 Cinematic - League of	League of Legends	Gaming	104159411	1.69
6	IN	2024-01-01	Protect \$500,000 Keep It!	MrBeast	Entertainment	85458562	4.21
7	JP	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
8	KR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	143955997	3.16
9	MX	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
10	US	2024-01-01	Grand Theft Auto VI Trailer 1	Rockstar Games	Gaming	166323421	6.73
11	BR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	126846652	3.54
12	CA	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	119170728	3.66
13	DE	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
14	FR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72

The results are sorted chronologically by month, then by the country giving monthly results.

- “category_title” having most distinct videos from 2022 and its percentage of the country’s total distinct videos

```

53 --4.) category_title having most distinct videos from 2022 and its percentage of the country's total distinct videos
54 WITH country_dist AS ( -- Calculate count of distinct videos for each country and category from 2022
55     SELECT country, trim(lower(category_title)) AS category_title, COUNT(DISTINCT video_id) AS total_category_video
56     FROM table_youtube_final
57     WHERE EXTRACT(YEAR FROM trending_date) >= 2022
58     AND category_title IS NOT NULL
59     GROUP BY country, category_title),
60 country_total AS ( -- Calculate total distinct videos for each country after 2022
61     SELECT country, COUNT(DISTINCT video_id) AS total_country_video
62     FROM table_youtube_final
63     WHERE EXTRACT(YEAR FROM trending_date) >= 2022
64     AND category_title IS NOT NULL
65     GROUP BY country),
66 category_percentage AS ( -- Calculate percentage and rank the categories by distinct videos grouping country
67     SELECT cd.country, cd.category_title, cd.total_category_video, ct.total_country_video, ROUND((cd.total_category_video * 100.0 /
68     ct.total_country_video), 2) AS percentage,
69     ROW_NUMBER() OVER (PARTITION BY cd.country ORDER BY cd.total_category_video DESC) AS rank_num
70     FROM country_dist cd
71     JOIN country_total ct ON cd.country = ct.country)
72 -- Select the top most categories for each country
73 SELECT country, category_title, total_category_video, total_country_video, percentage
74 FROM category_percentage
75 WHERE rank_num = 1
76 ORDER BY category_title, country;

```

This query determines which category has the most distinct videos per country since 2022, plus its percentage of the country's total. It calculates category totals, overall country totals, then calculates what is the percentage of category totals over the overall country's totals.

Result:

	COUNTRY	CATEGORY_TITLE	# TOTAL_CATEGORY_VIDEO	# TOTAL_COUNTRY_VIDEO	# PERCENTAGE
1	BR	entertainment	5417	23760	22.80
2	DE	entertainment	7709	30719	25.10
3	FR	entertainment	7548	32849	22.98
4	GB	entertainment	5643	27855	20.26
5	IN	entertainment	21281	50250	42.35
6	JP	entertainment	5658	17627	32.10
7	KR	entertainment	5122	15175	33.75
8	MX	entertainment	4195	17532	23.93
9	CA	gaming	6594	30869	21.36
10	US	gaming	6226	28799	21.62

The results display for very country, what category title has most distinct videos, majorly dominated by “Entertainment” and it also computes category totals, overall country totals and percentage.

- “channeltitle” producing most distinct videos: Vijay Television with 2049 distinct videos.

```

79
80 --5.) channeltitle producing most distinct videos
81 SELECT channeltitle, COUNT(DISTINCT video_id) AS distinct_videos
82 FROM table_youtube_final
83 GROUP BY channeltitle
84 ORDER BY distinct_videos DESC, channeltitle ASC
85 LIMIT 1;
86 --channeltitle = Vijay Television && distinct_videos = 2049
87
88 -----END-----
89

```

Results Chart

CHANNELTITLE	# DISTINCT_VIDEOS
Vijay Television	2049

This query identifies the channel with the most distinct uploaded videos overall. It groups data by “channeltitle” and counts distinct video IDs. The result displays only Vijay Television with the distinct video count of 2049.

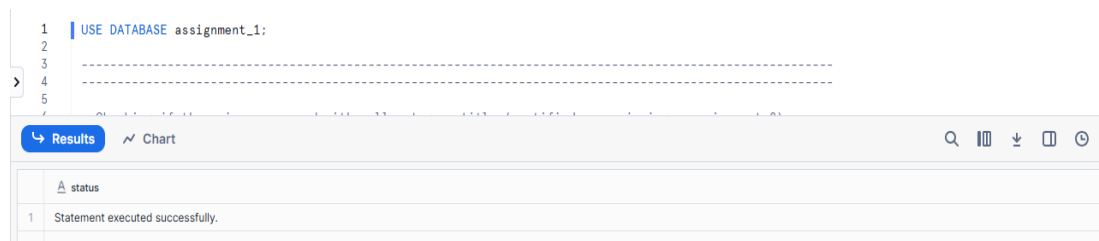
d. Part 4: Business Question

If there was a new YouTube channel launched tomorrow and the category was to be chosen, a data analyst would have taken some steps to analyse the YouTube Trending data and figured out the category of videos to be created, where “Music” and “Entertainment” categories are excluded. Need to make sure that the chosen strategy works in every country.

Introduction

Starting a new YouTube channel today requires more than just creativity, it needs the right choice of category to attract viewers. Since the “Music” and “Entertainment” categories are excluded, I analyzed other options using trending data from recent years. The goal was to identify which category consistently draws the most attention worldwide and whether it can work across different countries.

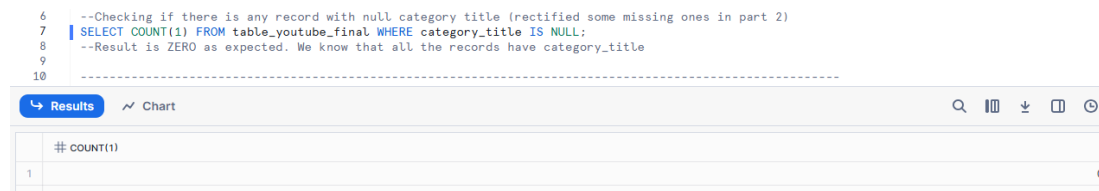
Database “assignment_1” was used. The table “table_youtube_final” is available here.



```
1 | USE DATABASE assignment_1;
2 |
3 | -----
4 | -----
5 |
```

status	
1	Statement executed successfully.

Checks for any records with null category title (it was rectified in part 2).



```
6 | --Checking if there is any record with null category title (rectified some missing ones in part 2)
7 | SELECT COUNT(1) FROM table_youtube_final WHERE category_title IS NULL;
8 | --Result is ZERO as expected. We know that all the records have category_title
9 |
10 | -----
```

#	COUNT(1)
1	0

Overall Performance of Categories

The three categories stand out well above the rest: Gaming category (472 billion views), People & Blogs category (404 billion views), and Sports category (347 billion views). These three categories attract far more viewers than areas like Comedy, Education, or Technology, which means they have stronger potential for growth if used as the focus for a new channel.

Checking the count of total views for all the categories:

```
13 --Checking the count of total view for all the categories
14 SELECT category_title, SUM(view_count) AS total_views
15 FROM table_youtube_final
16 WHERE category_title NOT IN ('Music','Entertainment')
17 GROUP BY category_title
18 ORDER BY total_views DESC;
```

Result:

	⚙ CATEGORY_TITLE	! # TOTAL_VIEWS
1	Gaming	472175892203
2	People & Blogs	404494421594
3	Sports	347228745414
4	Comedy	199787670068
5	Film & Animation	176560143845
6	Science & Technology	134528262116
7	News & Politics	85564342844
8	Howto & Style	64502920029
9	Education	64154614369
10	Autos & Vehicles	32609860598
11	Travel & Events	17438767290
12	Pets & Animals	14331489792
13	Nonprofits & Activism	2089504029

Viewership Across Countries

The next step was to check how these categories perform in various parts of the world. Gaming category is extremely popular everywhere. It ranks among the top categories in the US, UK, Mexico, Brazil, and Germany. People & Blogs category also does well and is stronger in countries like the US, Canada, and Mexico. Sports category has a huge fan base across the a few countries like US, UK, Canada, Germany, and India, but the type of sport followed could vary from country to country. According to the data, Gaming and Sports have shown more global dominance, while "People & Blogs" category performs well but leans toward specific regions.

Count of total view for all the categories for respective countries:

```
23 --Checking the count of total view for all the categories for respective countries as well
24 SELECT country, category_title, SUM(view_count) AS total_views
25 FROM table_youtube_final
26 WHERE category_title NOT IN ('Music','Entertainment')
27 GROUP BY country, category_title
28 ORDER BY country, total_views DESC;
```

Result:

	▲ COUNTRY	▲ CATEGORY_TITLE	## TOTAL_VIEWS
1	BR	Gaming	41923583045
2	BR	Sports	31624237825
3	BR	People & Blogs	26723156541
4	BR	Science & Technology	8538186750
5	BR	Comedy	8180745218
6	BR	Film & Animation	6493086188
7	BR	News & Politics	3107144506
8	BR	Education	2173339013
9	BR	Howto & Style	1335514155
10	BR	Autos & Vehicles	1005727831
11	BR	Travel & Events	584395801
12	BR	Pets & Animals	153210745
13	BR	Nonprofits & Activism	121882265
14	CA	Gaming	77839458463
15	CA	Sports	50000014550

Country-wise records for total views:

```
33 --Countrywise--
34 SELECT category_title, SUM(ctry_views) AS total_views,
35        LISTAGG(country || ' (= ' || TO_VARCHAR(ctry_views) || ' views)', ' || ' )
36        WITHIN GROUP (ORDER BY ctry_views DESC, country) AS countries_by_views
37 FROM (
38   SELECT category_title, country, SUM(view_count) AS ctry_views
39   FROM table_youtube_final
40   WHERE category_title IS NOT NULL
41   AND category_title NOT IN ('Music', 'Entertainment')
42   GROUP BY category_title, country
43 )
44 GROUP BY category_title
45 ORDER BY total_views DESC;
```

Result:

	▲ CATEGORY_TITLE	## TOTAL_VIEWS	▲ COUNTRIES_BY_VIEWS
1	Gaming	472175892203	US (= 86114934292 views) CA (= 77839458463 views) GB (= 71134953007 views) MX (= 61741932493 views) BR (= 41923583045 views) DE (= 403375
2	People & Blogs	404494421594	IN (= 90278954980 views) MX (= 50958043898 views) CA (= 46699356499 views) GB (= 43295521899 views) KR (= 41454832434 views) US (= 380372
3	Sports	347228745414	US (= 57233219021 views) GB (= 54141083999 views) CA (= 50908924559 views) DE (= 37227857994 views) MX (= 36542534281 views) BR (= 316242
4	Comedy	199787670068	IN (= 44769908310 views) CA (= 26131824399 views) KR (= 23298892089 views) US (= 20341872583 views) GB (= 19492139029 views) MX (= 148362
5	Film & Animation	176560143845	US (= 29256454223 views) IN (= 28397300707 views) CA (= 27933742765 views) GB (= 25849509996 views) MX (= 15737750544 views) DE (= 141840
6	Science & Technology	134528262116	CA (= 21951821199 views) US (= 21758148170 views) IN (= 21676198993 views) GB (= 19722794671 views) DE (= 15523590158 views) BR (= 8538186
7	News & Politics	85564342844	US (= 18254403823 views) KR (= 13308815804 views) CA (= 10827510638 views) GB (= 9082215261 views) IN (= 7925072531 views) DE (= 77600857
8	Howto & Style	64502920029	CA (= 11089840449 views) IN (= 10804284780 views) US (= 8465202810 views) GB (= 7215900829 views) KR (= 7051915273 views) MX (= 659452242
9	Education	64154614369	CA (= 12241958465 views) GB (= 11502266992 views) US (= 10625601655 views) DE (= 9391029879 views) IN (= 9165510238 views) KR (= 32727795
10	Autos & Vehicles	32609860598	GB (= 6147234802 views) CA (= 6116280726 views) US (= 5091086029 views) DE (= 4600441985 views) FR (= 2557603244 views) JP (= 1888040488 v
11	Travel & Events	17438767290	KR (= 5243658168 views) CA (= 2112340576 views) DE (= 2014694285 views) IN (= 1972153693 views) GB (= 1887641787 views) US (= 1520444554 v
12	Pets & Animals	14331489792	KR (= 4096029181 views) JP (= 3571626038 views) CA (= 1879509211 views) US (= 1581988423 views) GB (= 1329277571 views) IN (= 1005987842 v
13	Nonprofits & Activism	2089504029	KR (= 458327596 views) CA (= 362876349 views) US (= 278253528 views) GB (= 260193643 views) MX (= 244952510 views) JP (= 150024433 views)

Yearly Trends Viewership:

When looking at yearly trending data from 2020 to 2024, Gaming category does clearly dominate. In 2022 alone, it was the top category in eight different countries. Even though we have records available only till April of 2024, Gaming category dropped its lead in the first four months of 2024, it remained one of the strongest performers out of all others. “People & Blogs” category has also held steady in recent years, trending consistently in five countries in 2024. Sports category, on the other hand, has been growing at a good pace but is less consistent if compared to Gaming.

The chart comparing categories across years confirms this: Gaming category has been the most dependable winner, People & Blogs category has seen steady participation, and Sports category has shown growth but varies vastly depending upon the local interests.

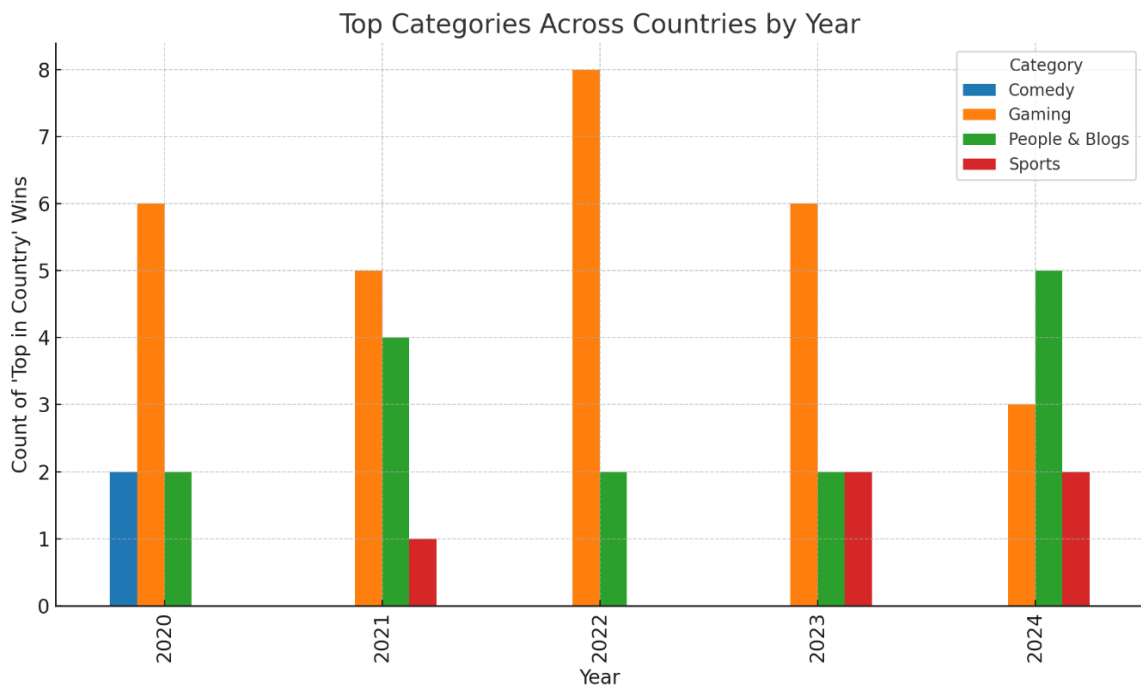
```
50 --List of trending category videos from 2020 to 2025
51 SELECT year, category_title, COUNT(*) AS leading_in_countries, SUM(total_views) AS sum_top_views
52 FROM (
53     SELECT year, country, category_title, total_views,
54           ROW_NUMBER() OVER (
55             PARTITION BY year, country
56             ORDER BY total_views DESC
57           ) AS rnk
58     FROM (
59         SELECT country, category_title, EXTRACT(YEAR FROM trending_date) AS year, SUM(view_count) AS total_views
60         FROM table_youtube_final
61         WHERE category_title NOT IN ('Music', 'Entertainment')
62               AND trending_date >= DATE '2020-01-01'
63               AND trending_date < DATE '2026-01-01'
64         GROUP BY country, category_title, year ))
65 WHERE rnk = 1
66 GROUP BY year, category_title
67 ORDER BY year, leading_in_countries DESC, sum_top_views DESC;
68
69 -----END-----
70
```

Result:

Results				
#	YEAR	CATEGORY_TITLE	# LEADING_IN_COUNTRIES	# SUM_TOP_VIEWS
1	2020	Gaming	6	35030617647
2	2020	Comedy	2	6432356812
3	2020	People & Blogs	2	4063918169
4	2021	Gaming	5	104008264722
5	2021	People & Blogs	4	58747247791
6	2021	Sports	1	5889373542
7	2022	Gaming	8	121808144752
8	2022	People & Blogs	2	59561419619
9	2023	Gaming	6	90227146387
10	2023	People & Blogs	2	26318459343
11	2023	Sports	2	16744285534
12	2024	People & Blogs	5	23691339812
13	2024	Gaming	3	15764024909
14	2024	Sports	2	5764176125

Global Fit

For a new channel, the ability to appeal to audiences worldwide is important. Gaming category has the clearest global reach, with strong numbers in multiple countries and a wide range of content ideas like playthroughs, reviews, tutorials, and live streams. People & Blogs category is also promising but may need to capture more regions to succeed everywhere. Sports category has good potential but is tied to specific events (like FIFA World Cup or NBA Finals, etc.) and regional preferences, which makes it less stable as a single focus.



Recommendation

The best choice of video category for a new YouTube channel must be Gaming category. It brings the highest total views, has been a leader across various countries year after year, and offers countless opportunities for engaging content and has proven its dominance across the seas. Viewers from different regions already enjoy Gaming content, which means the chances of building an international audience are much higher.

Considering diverse options available, if there is a plan to start a channel tomorrow, Gaming category would most definitely be the best and smartest option. It has proven its global popularity with consistent performance and a wide variety of content formats that can keep the audience, from vast countries, engaged for years to come.

Conclusion

The assignment helps to understand the practical use of Snowflake Data Lakehouse and Azure environment. As a data analyst, it is important to understand the data and it involves various steps like data ingestion, performing data cleaning, learning to read the data, performing in depth analysis to provide business requirements.

The dataset provided had all the top YouTube trending videos in ten countries. This dataset was successfully uploaded to the Azure environment and then injected the data into Snowflakes environment. Used SQL to clean the data, removed the duplicate values, updated missing values, and fetch various results as per business requirements.

While solving the business question, it was observed that “Gaming” category is highly recommended if a new YouTube channel is being launched considering the view counts over the period of 5 years and demand in 10 different countries (here categories like “Music” and “Entertainment” were excluded).

Overall, working with Snowflake and Azure has strengthened my confidence in solving business problems and handling large datasets, another milestone crossed in my journey into the world of data.

References

- Assignment Google Drive Link:
https://docs.google.com/document/d/1uMOXHb7I_3FgHqYELd8URxId4m4yE8Ei/edit?tab=t.0
- Kaggle link :<https://www.kaggle.com/rsrishav/youtube-trending-video-dataset>

